Paleomagnetic evidence for multiple Late Pliocene – Early Pleistocene glaciations in the Klondike area, Yukon Territory[1]

D.G. Froese, R.W. Barendregt, R.J. Enkin, and J. Baker

Abstract: The Late Pliocene – Early Pleistocene terraces of the Klondike area provide a conformal record of sedimentation which marks the transition from preglacial to glacial conditions, and is one of the most complete records of glaciation and interglaciation in the northern Cordillera. Preglacial sedimentation is recorded in the Lower White Channel gravel that contains a reverse-normal polarity sequence. A re-aggradation of the nonglaciated valleys of the goldfields in response to Late Pliocene cooling is recorded by the Upper White Channel gravel, which is characterized by a dominant normal polarity with a lower reversal likely associated with the first ice-wedge casts. Klondike gravel interfingers with Upper White Channel and is normally magnetized, indicating an equivalent magnetostratigraphic unit. This glaciofluvial gravel provides evidence of extra-basinal clasts, marking a major reorganization of drainage associated with the first advance of the Cordilleran Ice Sheet in western Yukon Territory correlated with the Gauss chron (>2.58 Ma). An intermediate terrace in Klondike valley is overlain by 8 m of fluvial and glaciofluvial gravel named the Midnight Dome gravel. This gravel is mantled by 15 m of loess and hillslope deposits, named the Midnight Dome loess, which preserve at least three interglacial pollen assemblages, and a reverse-normal-reverse-normal polarity sequence assigned to the late Matuyama chron, including Jaramillo subchron (1.07–0.99 Ma), and early Brunhes chron (<0.78 Ma). The oldest glaciations here, based on the preferred interpretation, extend well into the Pliocene (>2.58 Ma) making these among the oldest glacial sediments in North America.

Résumé : Les terrasses du Pliocène tardif – Pléistocène précoce de la région du Klondike fournissent un registre concordant de la sédimentation qui marque la transition de conditions préglaciaires à glaciaires; c'est l'un des registres les plus complets sur la glaciation et l'interglaciation de la Cordillère nord. La sédimentation préglaciaire est révélée dans les graviers du chenal White inférieur qui contient une séquence de polarité inverse normale. Un nouveau nivelage du terrain par déposition dans les vallées non glaciaires des districts aurifères, en réaction au refroidissement du Pliocène tardif, est révélé dans le gravier du chenal White supérieur, lequel est caractérisé par une prédominance de la polarité normale et présente une inversion moindre probablement associée aux premières fentes de glace fossiles. Le gravier du Klondike s'entrelace avec celui du chenal White supérieur et il est normalement magnétisé, indiquant une unité magnétostratigraphique équivalente. Ce gravier fluvio-glaciaire fournit des évidences de clastes hors bassin, indiquant une réorganisation majeure du drainage associée à la première avancée de l'Inlandsis de la Cordillère dans l'ouest du Territoire du Yukon en corrélation avec le chron de Gauss (>2,58 Ma). Une terrasse intermédiaire dans la vallée du Klondike est recouverte de 8 m de gravier glaciaire et fluvio-glaciaire que l'on nomme le gravier Midnight Dome. Ce gravier est recouvert de 15 m de lœss et de dépôts de flanc de coteaux, le lœss Midnight Dome, qui conservent au moins trois assemblages de pollen interglaciaire et une séquence de polarité inverse - normale - inverse - normale assignée au chron Matuyama tardif, incluant le sous-chron Jaramillo (1,07–0,99 Ma), et au chron Brunhes précoce (<0,78 Ma). Ici, selon l'interprétation préférentielle, les plus anciennes glaciations s'étendent loin dans le Pliocène (>2,58 Ma) ce qui les classe parmi les plus anciens sédiments glaciaires en Amérique du Nord.

[Traduit par la Rédaction]

Received April 2, 1999. Accepted December 23, 1999.

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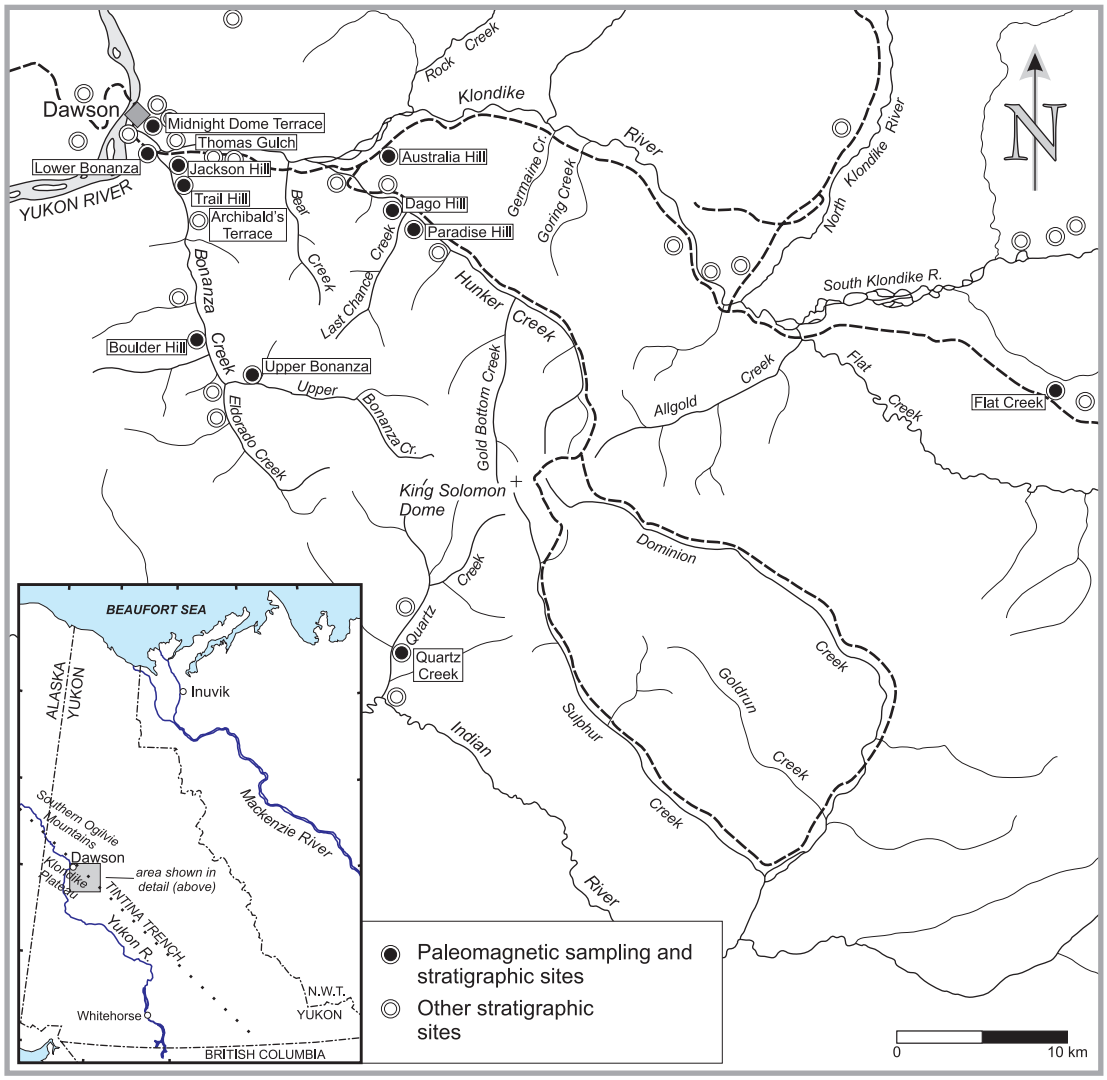
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Introduction

For nearly a century, the lower Klondike valley and its gold-bearing tributaries have been recognized for their exceptional preservation and exposures of Pliocene and Pleistocene deposits (McConnell 1905, 1907). The pioneering work of R.G. McConnell remains the most complete description, and with only minor changes, the basic framework for the alluvial deposits of the Klondike goldfields. Subsequent research on the Klondike has focussed on the geomorphic setting (Hughes et al. 1972; Milner 1977), depositional environments (Morison and Hein 1987; Fraser

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Fig. 1. Klondike valley study area and sites described in text.



and Burn 1997), characteristics of hydrothermal alteration (Tempelman-Kluit 1982; Dufresne et al. 1986), Pleistocene vertebrate fossils (Harington and Clulow 1973), and the connection between lode and placer deposits (Gleeson 1970; Knight et al. 1994). Yet, despite the considerable economic resources of these sediments, and their recognized link to Late Cenozoic environments, chronology for the alluvial deposits is not well established.

The purposes of this study are to (1) provide a chronology of the terraces with reference to the geomagnetic polarity time scale; and (2) determine the paleoclimatic significance of these deposits. In this paper, we present observations on the depositional environments and magnetic polarity of preglacial, and multiple glacial and interglacial deposits in the Klondike goldfields during the Late Pliocene and Pleistocene.

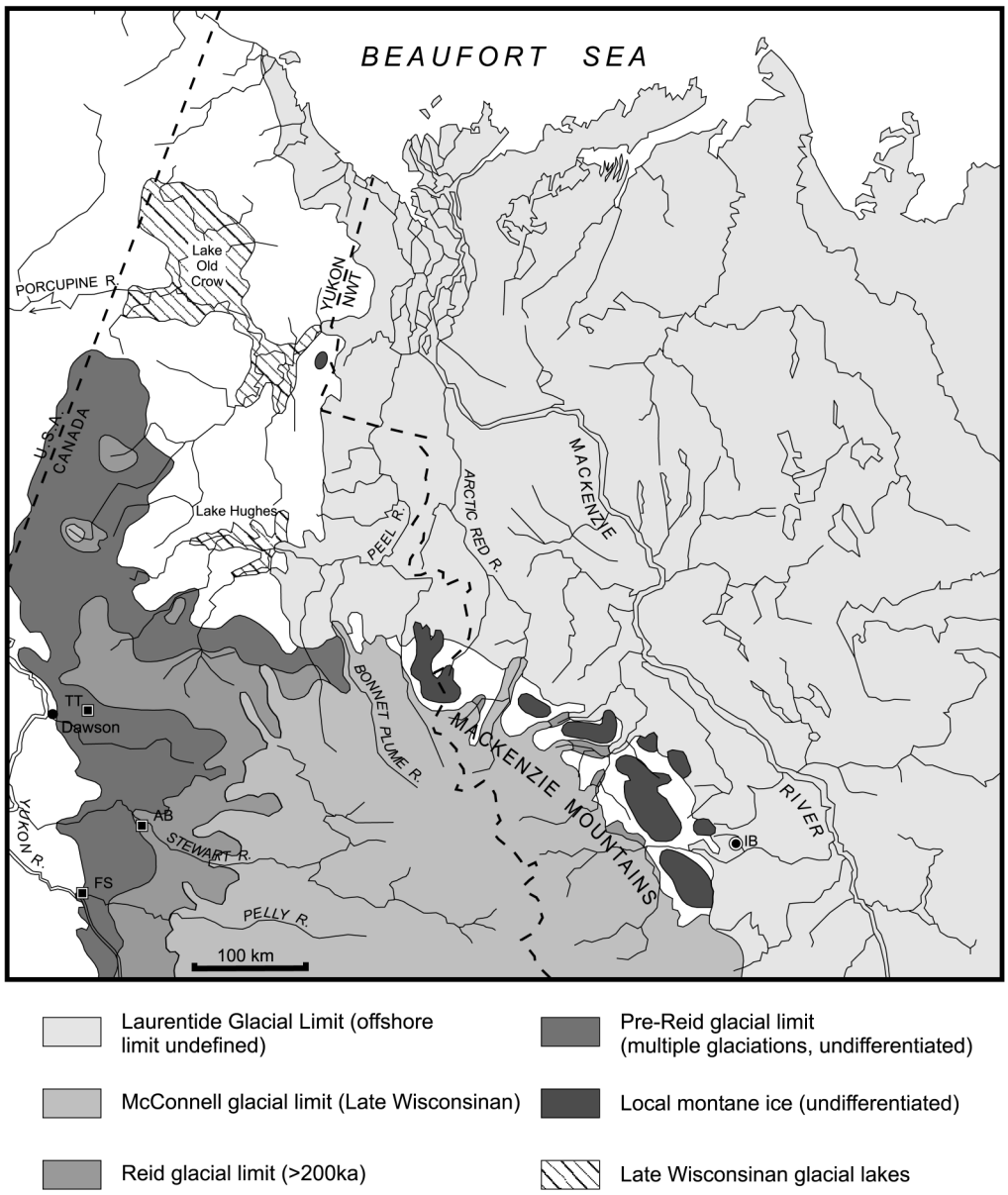
Regional setting

The lower Klondike River valley is west of the confluence of the North and South Klondike rivers, and continues to the Yukon River at Dawson City (Fig. 1). The high-level terraces that are the focus of this study are located throughout this area, but exposures are generally limited to the area of recent mining, from approximately Hunker Creek to Dawson City (Fig. 1).

The Klondike area is within the unglaciated region of western Yukon Territory. A major physiographic feature in Yukon, the Tintina Trench, is located immediately north of the Klondike area. This trench is a Late Tertiary graben which locally separates Klondike Plateau from Southern Ogilvie Mountains (Tempelman-Kluit 1980). To the north, Southern Ogilvie Mountains include peaks over 2000 m, and provided a centre for local ice masses during the Pleistocene. South of Tintina Trench, Klondike Plateau is a largely unglaciated, gently sloping upland consisting of concordant summits, locally termed "Domes," connected by ridges separated by deeply incised, V-shaped valleys. The Plateau is considered to be an uplifted erosional surface produced as a result of extensive subaerial exposure in the early to mid-Tertiary (McConnell 1905; Tempelman-Kluit 1980). Tempelman-Kluit (1980) hypothesized that, during this time, a south-flowing paleo-Yukon River existed, connecting the area north of Tintina Trench with the Pacific Ocean. He ar-

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Fig. 2. Generalized map of glacial limits for northwestern Canada and adjacent Alaska (from Duk-Rodkin et al. 1996). Locations of multiple glaciation sections providing Late Pliocene to mid- Pleistocene chronologies for Yukon – Northwest Territories discussed in text: AB, Ash Bend; FS, Fort Selkirk; IB, Inlin Brook; TT, Tintina Trench; Liard River sections outside diagram to southeast.



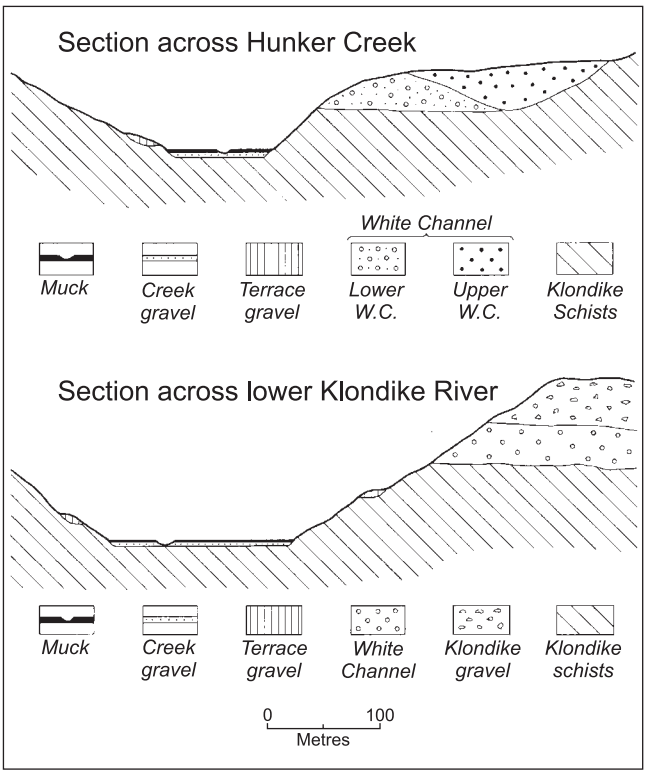
gued that southerly drainage of Yukon Plateau may have persisted through the uplift of ranges in southwestern Yukon, and was likely diverted to the northwest by the growth of late Cenozoic ice sheets.

Quaternary history

Bostock (1966) proposed a record of four successively less extensive glaciations of the Cordilleran Ice Sheet in central Yukon, which, from oldest to youngest, he named the Nansen, Klaza, Reid, and McConnell. McConnell-age deposits correlate to Late Wisconsinan glaciation (Hughes et al. 1989; Matthews et al. 1990), while Reid-age deposits occur below the Sheep Creek tephra (ca. 200 ka) at the Ash Bend site (Hughes et al. 1987; Berger et al. 1996; Fig. 2). More recently, multiple Late Pliocene to mid-Pleistocene advances have been recognized and grouped as "pre-Reid" (Hughes et al. 1989). From the Liard River area of southeastern Yukon, Klassen (1987) described two pre-Reid events: till A, below basalts >765 ka, and till B prior to 232 ka. In the Fort Selkirk area, Jackson et al. (1996) reported two late Matuyama-age (reversed magnetic polarity) pre-Reid glacial events (between 1.77 and 0.78 Ma). Hughes et al. (1993), Barendregt et al. (1996) and Duk-Rodkin et al. (1996) presented evidence for at least four pre-Reid glaciations in Mackenzie Mountains, eastern Cordillera, of which the earliest occurred within the Matuyama chron (reversely magnetized). In the Stewart River area, Bond (1997) reported a record of at least two pre-Reid glaciations. In addition to Cordilleran Ice Sheet advances, local glaciation of the Southern Ogilvie Mountains occurred. Vernon and Hughes (1966) noted at least three advances that have been correlated on the basis of relative preservation and soil de-

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Fig. 3. Generalized sections across terrace gravels of Hunker Creek and Klondike River, Yukon (after McConnell 1907).



velopment with pre-Reid, Reid, and McConnell events (Tarnocai et al. 1985; Duk-Rodkin 1996).

Klondike goldfields stratigraphy

McConnell (1905, 1907) divided the gravel of the Klondike goldfields into stream and gulch gravel, terrace deposits, and high-level terraces, which included the White Channel and a high-level river gravel, known as the Klondike gravel (Hughes et al. 1972; Morison and Hein 1987). These relations are shown in Fig. 3, and lithostratigraphic logs from sites described during fieldwork in 1995–1998 are shown in Fig. 4.

Lower White Channel gravel

The Lower White Channel is a characteristically light grey to white quartz-rich gravel that, at some sites, has been strongly leached by hydrothermal fluids (Tempelman-Kluit 1982; Dufresne et al. 1986). Lithostratigraphic logs from Australia, Jackson, Paradise, and Dago hill sites include Lower White Channel sediments. Lithologies consist mainly of white quartz, quartz–muscovite–chlorite schist and volcanic rocks derived entirely from the local drainage basin. The deposit is found on terraces cut into bedrock along Bonanza and Hunker creeks and has been interpreted as a braided-river deposit derived from preglacial uplift and weathering of the drainage basin (Morison and Hein 1987).

Upper White Channel gravel

The Upper White Channel was observed at Dago, Paradise, Australia, and Jackson hills and Lower Bonanza sections (Fig. 4). The deposit is entirely derived from local lithologies, and clasts are unleached and better preserved than those found in the Lower White Channel. Ice-wedge casts and involutions in the sediments may indicate the first evidence of permafrost in the Klondike area (Fig. 5*a*). At Jackson Hill, a 60 cm deep ice-wedge cast is developed near the contact with the Klondike gravel (Fig. 5*b*). At other sites, Paradise Hill, Dago Hill, and Quartz Creek, ice-wedge casts have been observed (six in total developed in gravel) consistent with previous observations of cryoturbation in the White Channel (Hughes et al. 1972; Morison 1985). Sedimentologically, the Upper White Channel gravel is interpreted as a braided stream deposit that aggraded under periglacial conditions. At the Dago, Paradise, and Australia hill sites, there is a prominent unconformity between the Upper White Channel and altered Lower White Channel gravel, marked by up to 12 m of aggradation of the Upper White Channel (Fig. 6). It represents the last aggradation of the White Channel during Late Pliocene cooling, and the top of the unit interbeds in the lower valleys of Bonanza and Hunker creeks with Klondike gravel (Hughes et al. 1972).

Klondike gravel

Klondike gravel is lithologically distinct from the locally derived White Channel gravel. The presence of chert, various quartzites, and slate clasts indicate a source in Selwyn continental-margin sedimentary rocks, found north of Tintina Trench. These rocks were transported by glacial meltwater into Klondike River valley, providing the first evidence of glaciation in the region (Hughes et al. 1972). The areal extent of this gravel spreads eastward to the Flat Creek beds (McConnell 1905; Fig. 1) and westward along the Yukon River to at least the Alaska border (Duk-Rodkin 1996).

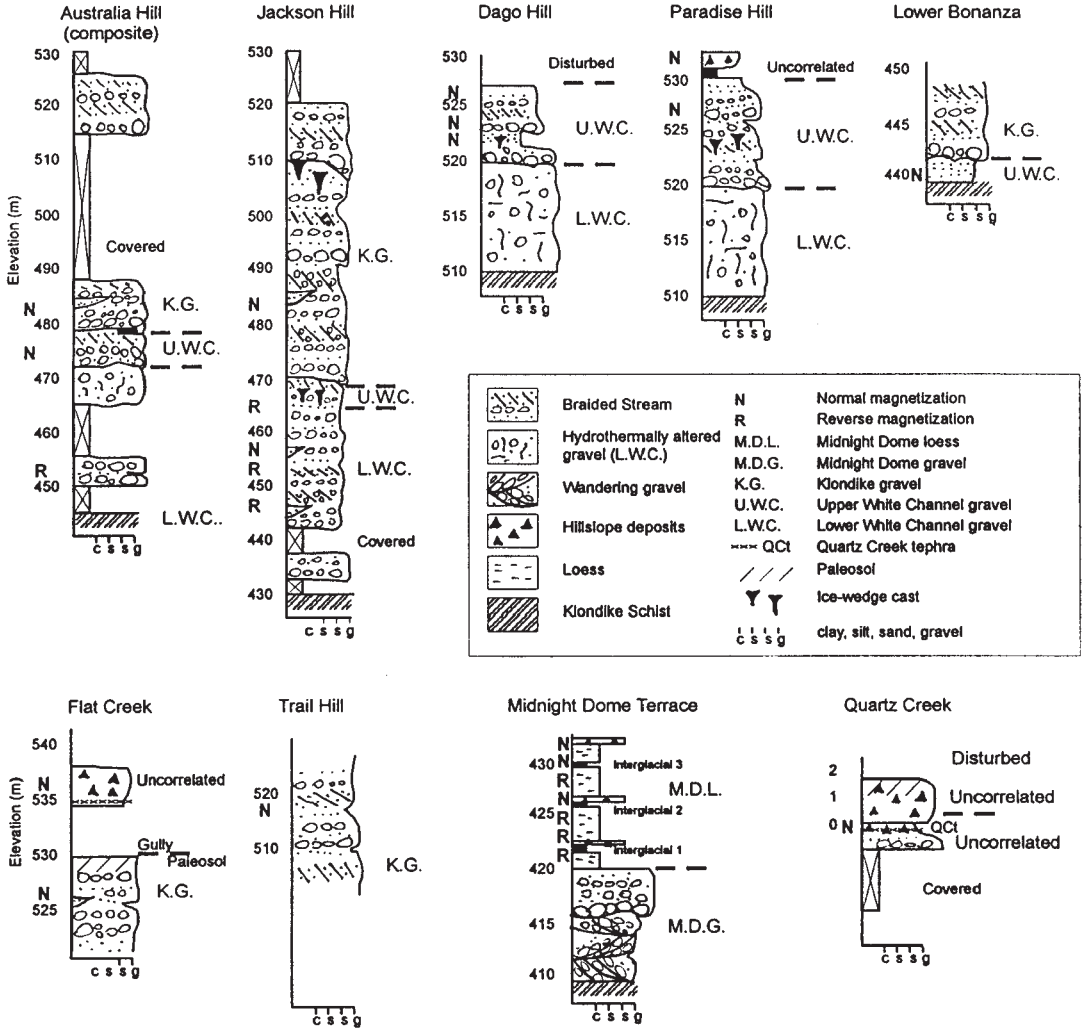
Klondike gravel (Fig. 7) consists of braided-stream deposits, which are largely massive and imbricate at Flat Creek, and show a greater frequency of planar-tabular crossbeds in the more westerly Klondike valley sites at Trail, Jackson, and Australia hills (Fig. 4), suggesting a more distal position from the paleo-ice margin (Smith 1985). Paleoflow measurements at Flat Creek indicate a source from the southeast, suggesting the Klondike gravel is derived from the first Cordilleran Ice Sheet advance into western Yukon, rather than local glaciation of the Ogilvie Mountains to the north (Bond 1997).

Intermediate terraces

Terraces intermediate in elevation between the White Channel – Klondike gravel level and present valley bottom are poorly preserved in the Klondike area. Milner (1977) and Naeser et al. (1982) investigated Archibald's Terrace (Fig. 1) which includes the Mosquito Gulch tephra on a terrace incised at least 60 m into the White Channel level. The fission track age of 1.22 Ma (minimum age) indicates a clear pre-Reid age for the surface. One intermediate terrace on the Midnight Dome, incised 100 m into the White Channel – Klondike gravel, was investigated in detail. The Midnight Dome site has previously only been reported with respect to the gold-bearing gravel at the site (Froese and Hein 1996). Its age and relation to Archibald's Terrace will be discussed later in reference to the paleomagnetic results.

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Fig. 4. Lithostratigraphic logs (generalized) at sites where paleomagnetic samples were collected and yielded characteristic polarity (bold N, R on left side of columns, normal and reverse magnetization, respectively). See Fig. 1 for site locations.



Midnight Dome gravel

The Midnight Dome gravel, a unit introduced here, consists of two successive depositional environments. Overlying bedrock is an interglacial "wandering gravel bed" river deposit (Desloges and Church 1987) similar to the modern Klondike River, dominated by low-angle lateral accretion facies (Froese and Hein 1996; Froese 1997). This is overlain by 2–5 m of coarse, cobble-gravel interpreted as a proximal braided deposit. This aggradation represents an outwash deposit as a result of a pre-Reid glaciation in Southern Ogilvie Mountains (Froese and Hein 1996; Fig. 8). Subsequent incision of the terrace by the paleo-Klondike River resulted in a transition to dominantly aeolian-colluvial deposition on the terrace surface (Figs. 4, 8).

Midnight Dome loess

The Midnight Dome loess, a unit previously unrecognized, overlies the Midnight Dome gravel and is up to 15 m thick. The deposit consists of massive and retransported loess units 1–4 m thick separated by at least three laterally continuous organic silt horizons (5–30 cm thick) containing interglacial pollen assemblages dominated by *Picea* and *Abies* forest taxa (Schweger et al. 1999). Interglacial 1 occurs approximately 1 m above the Midnight Dome gravel and can be traced laterally across the exposed section >100 m. Interglacial 2 is defined from pollen collected within a retransported organic silt up to 10 cm thick, within a major colluvial diamict unit midway through the section. This organic horizon is found in wavy, discontinuous lenses up to 10 cm thick within the gravelly diamict unit and was repeated in two transects. Interglacial 3 is found 1 m above interglacial 2 and has a maximum thickness of 30 cm and was similarly sampled in two transects. The age of these units will be discussed later with respect to the paleomagnetic results.

Reid and McConnell deposits

Younger terraces are preserved east of Dawson City, near the confluence of the North and South Klondike rivers. Probable "Reid-age" glaciofluvial gravel is present 40 m above the floodplain at the confluence (Tarnocai et al. 1985; Duk-Rodkin 1996), while the McConnell-age terrace surface

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Fig. 5. Sedimentary structures interpreted as ice-wedge casts developed in (*a*) earliest glacial Klondike gravel, midway through valley fill at Jackson Hill, and (*b*) preglacial Upper White Channel gravel near contact with Klondike gravel at Jackson Hill. Ice axe is 85 cm long.





grades into the North Klondike River above the confluence, and thus no recognizable terrace is found (Duk-Rodkin 1996).

Paleomagnetism

Oriented samples were collected within fine-grained, overbank sediments in gravel deposits, while loess deposits were sampled with a minimum vertical interval of 15 cm. Where contacts with adjacent facies were sharp, as in the case of overbank and some eolian–colluvial deposits, additional samples were collected horizontally to obtain at least 7 samples. Sediments were collected by cleaning the exposure to a vertical face and inserting plastic cylinders (2.5 cm diameter) horizontally. Sample azimuths were measured using a magnetic compass. Remanence measurements were made on an AGICO JR-5A spinner magnetometer. Stepwise alternating field demagnetization was carried out using a Schonstedt GSD-5 with tumbler in peak fields up to 100 mT. Samples were demagnetized using 5–10 steps and directions determined by principal component analysis (Kirschvink, 1980).

Since the sediments are flat-lying, there is no difference in direction between a primary normal polarity remanence and a present field overprint. Thus, units containing reverse polarity are more readily characterized than those of normal polarity. However, in some samples, we have evidence that a pre-Brunhes age normal polarity magnetization is retained. The Klondike gravel at Flat Creek has three magnetic remanence components: a low stability normal polarity, an intermediate stability reverse polarity, and a highly stable normal polarity (Fig. 9*d*). We interpret the first normal to be a present field overprint, the reverse to be a Matuyama-age partial remagnetization over the older normal polarity high stability primary remanence of Olduvai or Gauss age.

Directions sometimes differ from the expected dipole field by more than 30°, but these deviations do not invalidate a polarity assignment. In particular, reverse polarity directions can be affected by large normal present field overprints which are incompletely removed during demagnetization. One often finds coherent directions from a single horizon, but considerable differences between horizons in a single unit. This inter-horizon variability and intra-horizon consistency indicates that paleosecular variation was recorded, and provides confidence that the polarity at the time of deposition was retained.

In the highly micaceous Lower and Upper White Channel gravel, we observed a strong correlation between the sample insertion direction (sampling azimuth) and the natural remanent magnetization (NRM) declination. This sampling insertion overprint masks the primary direction in some samples, but is not so strong as to affect the polarity assignment.

Natural remanent magnetization intensities ranged from 0.3 to 271 $mA \cdot m^{-1}$, with the Pliocene White Channel gravel having weaker magnetization than most Pleistocene sediments. Median destructive fields range from 10 to

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Fig. 6. Dago Hill, lower Hunker Creek (Fig. 1), shows the relationship between the Upper and Lower White Channel gravel. At this site, and the adjacent high terrace exposures between Paradise and Australia hills, a distinct reaggradation of the valley is marked by a sharp boundary above altered Lower White Channel (primary sedimentary structures not distinguishable) and the Upper White Channel gravel which is undisturbed.

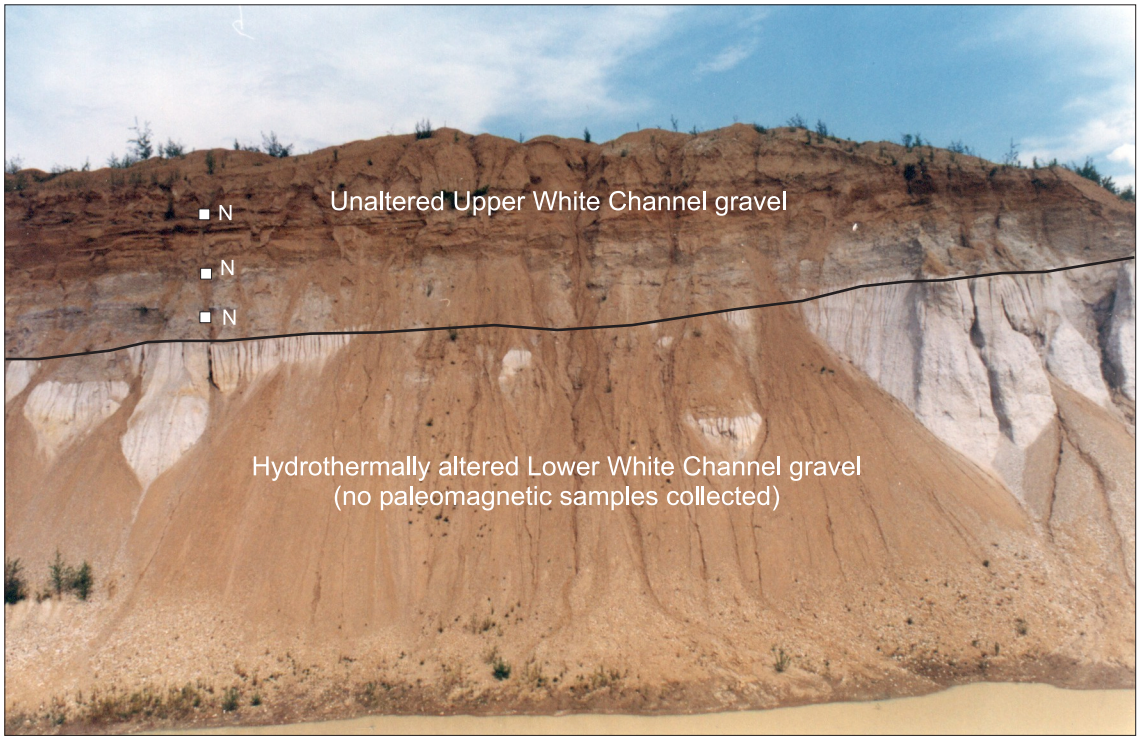
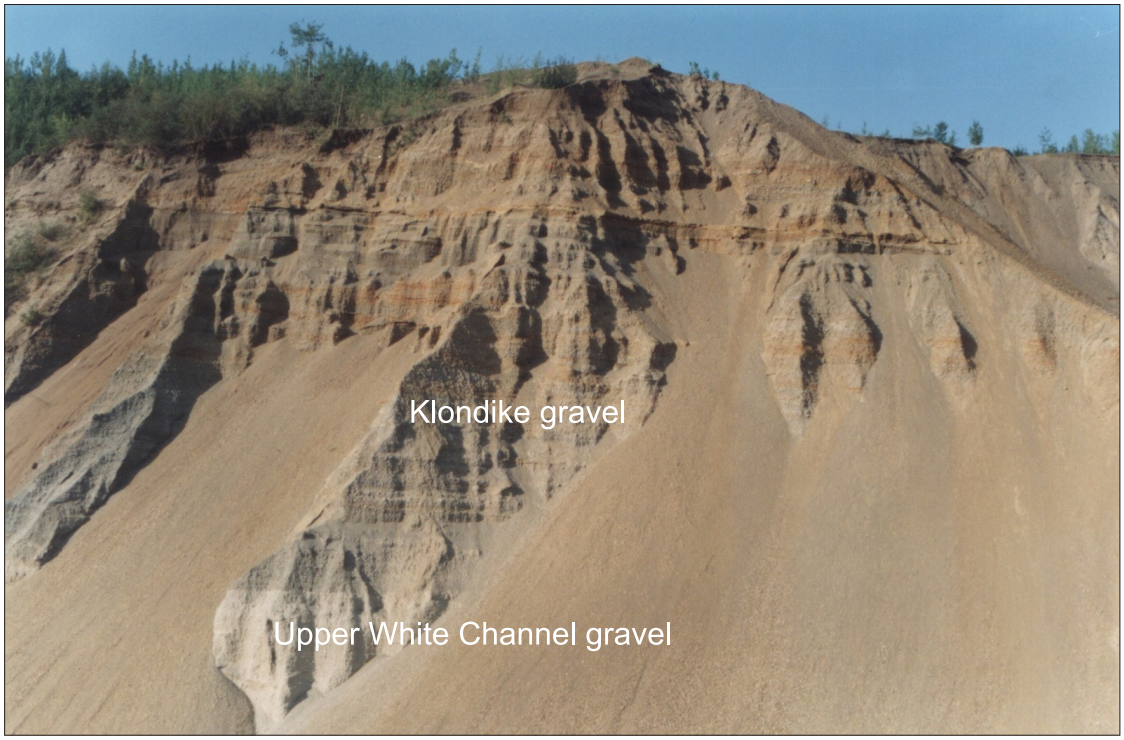
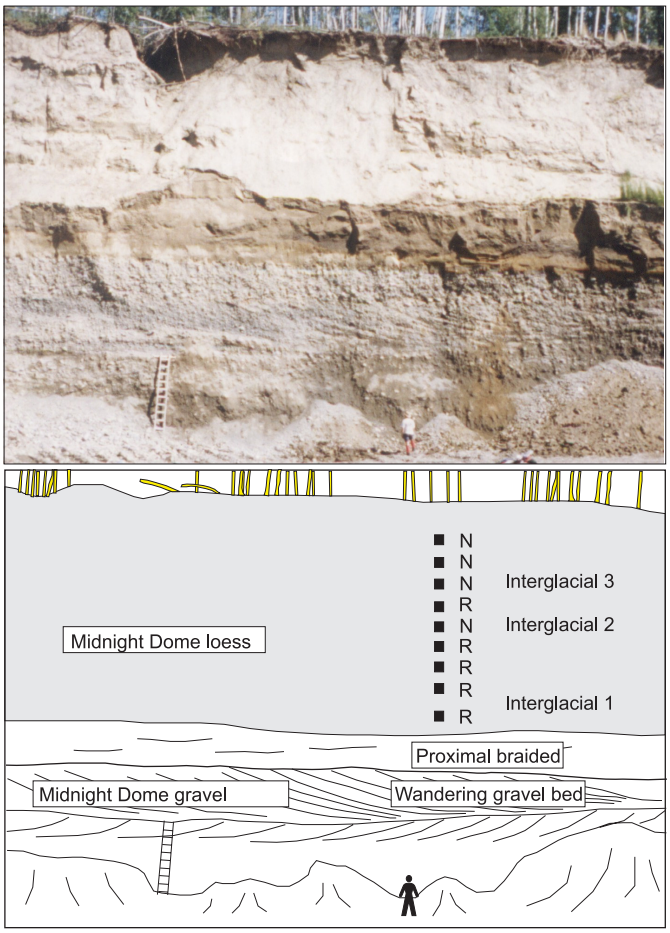


Fig. 7. Jackson Hill, located on lower Bonanza Creek (Fig. 1), consists of the Klondike gravel (upper 40 m) which overlies White Channel gravel (upper 10 m exposed). The Klondike gravel interfingers with the Upper White Channel along Bonanza and Hunker creeks in the more lateral braidplain positions, and marks the first direct evidence of glacial conditions in the region in the Late Pliocene.



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Fig. 8. The lower Midnight Dome gravel consists of laterally accreting low-angle planar-crossbeds of a paleo-wandering gravel bed river, overlain by 2–5 m of coarse, cobble-gravel interpreted as a proximal braided deposit (Froese and Hein 1996). The Midnight Dome gravel is overlain by up to 15 m of glacially derived loess and hillslope deposits (Midnight Dome loess) of mixed polarity (R-N-R-N), which contains at least 3 interglacial pollen assemblages (Schweger et al. 1999).



60 milliTesla (mT). Greater confidence is assigned to those samples and sampling horizons having median destructive fields of 40–60 mT. Typical demagnetization curves and their explanations are given in Fig. 9. In most samples, a single component of magnetization was observed during demagnetization (Figs. 9*a*, 9*b*), but, for a few samples, a more complex magnetization was measured (Fig. 9*c*). A summary of the mean remanent magnetization directions for each unit after alternating field cleaning is given in Table 1. Plots of mean directions and associated 95% confidence intervals (α_{95}) are shown in Fig. 10.

Summary of magnetostratigraphy

Table 1 provides a summary of sampling sites and polarity data by site and stratigraphic unit. Lithostratigraphic logs of sections and sampling locations are shown graphically in Fig. 4. More detailed descriptions of sites and remanence characteristics of individual samples are reported in Froese (1997).

Table 1. Summary of paleomagnetic remanence directions.

Unit	*n*	*D* (°)	*I* (°)	*k*	α_{95} (°)	Polarity
Midnight Dome loess (elevations (m) above top of Midnight Dome gravel)						
1 (8.9–12 m)	25	47	65	21	7	N
2 (8.3–8.9 m)	5	123	–73	27	15	R
3 (6.0–8.3 m)	18	12	69	20	8	N
4 (1.2–6.0 m)	26	191	–67	12	9	R
5 (1.0–1.2 m)	9	—	—	—	—	Incoherent
6 (0.8–1.0 m)	5	149	–53	7	32	R
7 (0.0–0.8 m)	21	123	–69	9	12	R
Klondike gravel						
1Australia Hill	21	17	80	26	6	N
2 Jackson Hill	6	16	68	22	15	N
3 Trail Hill	6	71	70	25	14	N
4 Flat Creek	8	31	52	36	11	N
5 Flat Creek	7	166	–32	3	40	R Overprint
Upper White Channel						
1 Quartz	4	330	79	22	20	N
2 Australia Hill	13	24	81	8	16	N
3 Paradise Hill	17	*247	75	27	7	N
4 Dago Hill	14	16	71	69	5	N
5 Lower Bonanza	5	51	76	73	9	N
6 Jackson Hill	5	*311	–75	7	32	R
Lower White Channel						
1 Boulder Hill	13	313	73	48	6	N
2 Jackson Hill	10	9	81	32	9	N
3 Australia Hill	15	151	–56	17	9	R

Note: *n*, number of samples; *D* and *I*, declination and inclination of the mean remanence direction; *k*, precision parameter; α_{95}, circle of confidence ($P = 0.05$); * sampling insertion overprint affected declination, inclination unaffected (see text); *N*, normal magnetization, *R*, reverse magnetization.

Lower White Channel gravel

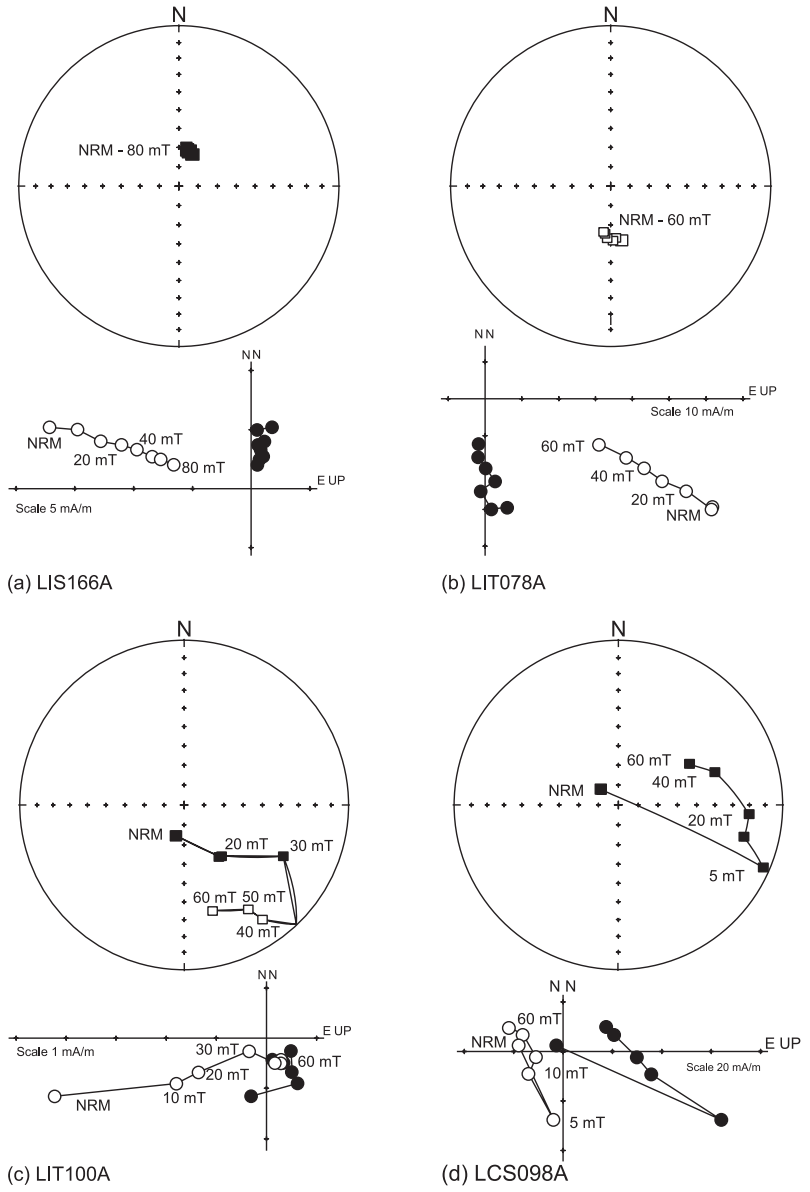
Paleomagnetic samples from Australia Hill, collected near the bedrock contact, indicate deposition began during a reversely magnetized interval. Additional samples from Jackson Hill, between 20 and 40 m above bedrock, indicate a reverse-normal polarity sequence. Samples from Boulder Hill are normally magnetized and presumably represent a normal consistent with either the lower or upper Jackson Hill normal. Thus, in summary, normal and reverse polarity is recognized in the Lower White Channel gravel.

Upper White Channel gravel

At Jackson Hill, reversely magnetized sediments occurring above a probable ice-wedge cast (Fig. 5*b*) record a short-lived reversal (probably a subchron) within the Upper White Channel. This reversed polarity was not found at the Hunker Creek Upper White Channel sites where ice-wedge casts have been noted, but are normally magnetized. Given the size of the ice-wedge cast at Jackson Hill, 50–60 cm depth, some caution should be maintained for this earliest evidence of permafrost. The Hunker Creek sampling sites

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Fig. 9. Representative demagnetization characteristics of the collection. For each sample, the stereograph plots magnetization direction at each step, where the filled (open) squares lie on the lower (upper) hemisphere. The orthogonal plot shows the horizontal (vertical) projection of each step with a closed (open) circle. (*a*) and (*b*) show simple mono-component demagnetization with normal and reverse polarity, typical of most of the collection. The reverse polarity characteristic direction of (*c*) has a normal polarity overprint. In (*d*), the complex magnetization recorded in the Klondike gravel at Flat Creek site consists of three magnetic remanence components: a low stability normal polarity, an intermediate stability reverse polarity, and a highly stable normal polarity. We interpret the first normal to be a present field overprint, the reverse to be a Matuyama-age overprint and the high stability normal to be the primary remanence of pre-Brunhes age (Olduvai or Gauss normal).



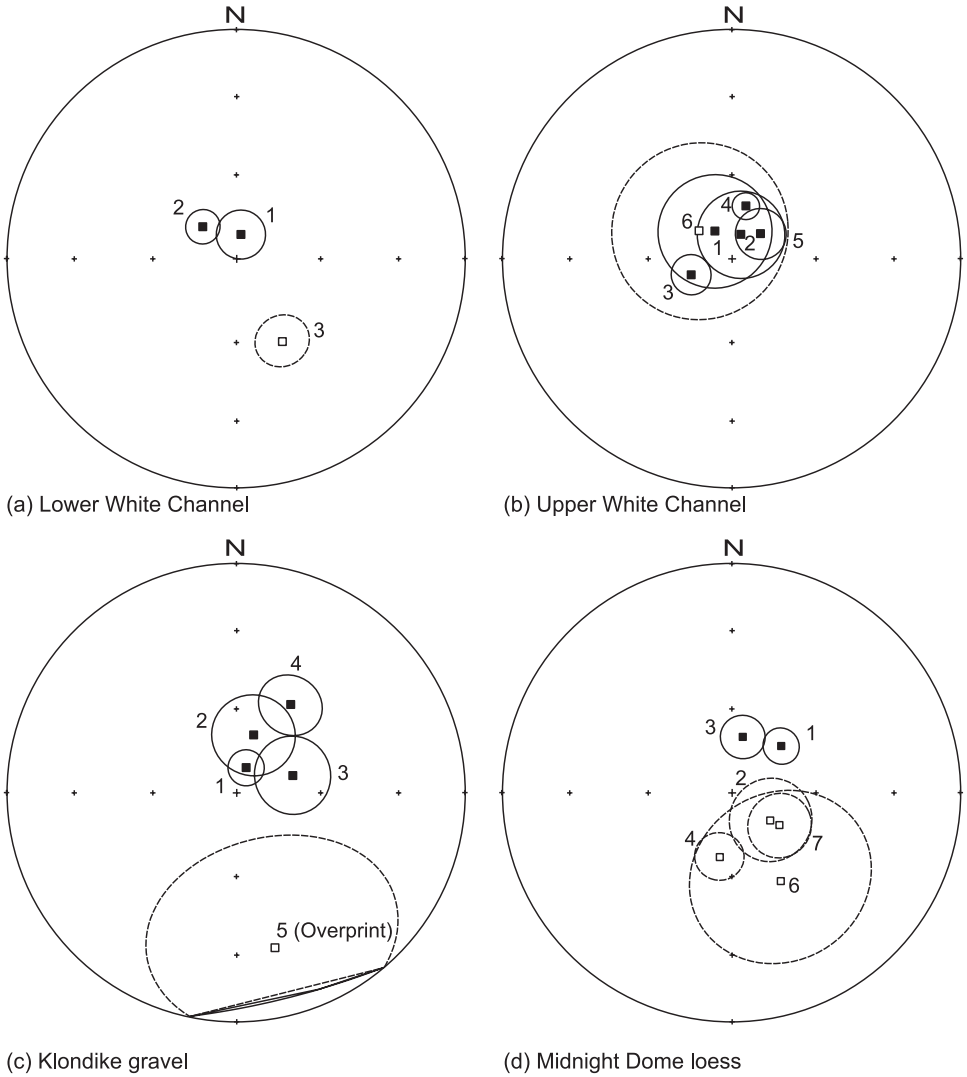
(Paradise Hill, Dago Hill, Australia Hill) all show normal magnetization above the unconformity and interbed with the Klondike gravel between Australia and Dago hills (Fig. 1). One site south of King Solomon Dome, at Quartz Creek (Fig. 1), is also normally magnetized at the location of the Quartz Creek tephra (Kunk 1995). The range in ages on the Quartz Creek tephra, from 2.6–3.1 Ma ($^{40}Ar/^{39}Ar$: Kunk 1995) to >0.8 Ma (fission track: N.D. Naeser, personal communication *in* Morison et al. 1998) may be consistent with the normal magnetization at the site (either Gauss or a Matuyama subchron). However, the distance from the remaining sites investigated in this study and the lack of any systematic work defining White Channel gravel south of King Solomon Dome make it difficult to place stratigraphic significance to either age.

Klondike gravel

Paleomagnetic samples were collected from four sites, and a consistent normal polarity was determined at each (Table 1). At Flat Creek, a reversed overprint (intermediate stability) was isolated from a high stability primary normal, indicating a pre-Brunhes normal magnetization (either a

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Fig. 10. Stereographic plots of unit means with their associated 95% confidence intervals. The lower or normal polarity (upper or reverse polarity) hemisphere is represented by closed (open) squares and solid (dashed) lines. The number labels correspond to units listed in Table 1.



Matuyama subchron or Gauss chron; Fig. 9*d*). In lower Klondike River valley, Klondike gravel interfingers with Upper White Channel and on the basis of a common normal polarity with the Hunker Creek sites, is considered a contemporaneous magnetostratigraphic zone. Thus, the onset of glacial conditions in central Yukon, recorded by the Klondike gravel, occurred during a pre-Brunhes normal chron or subchron.

Midnight Dome loess

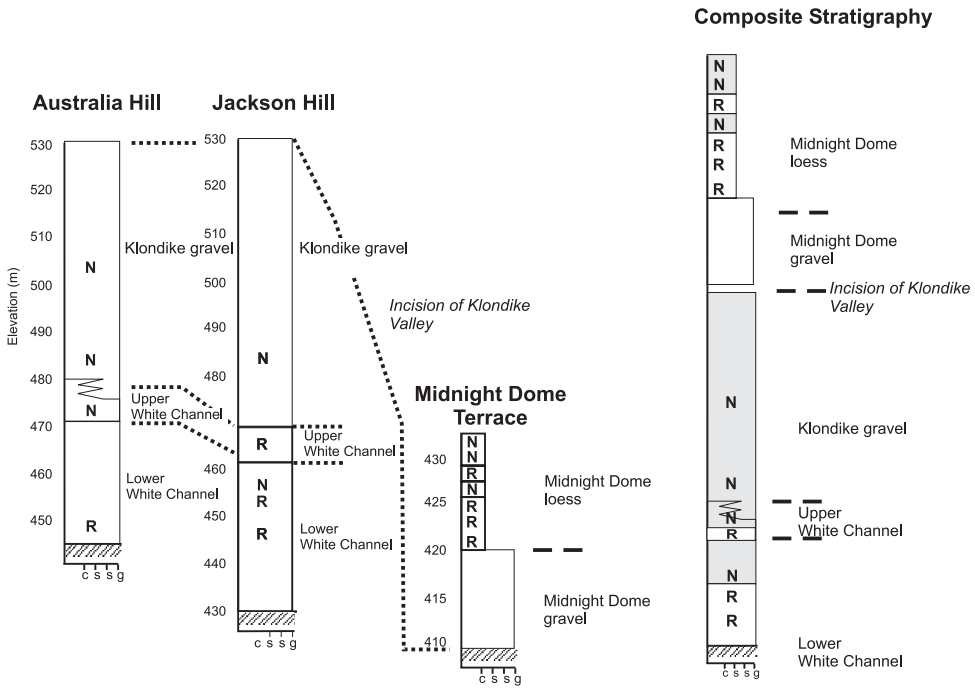
The Midnight Dome loess was the youngest unit sampled for paleomagnetic analysis. Paleomagnetic results from the loess, reworked loess, and organic silts indicate a reverse-normal-reverse-normal magnetic sequence which was repeated in samples from two transects 200 m apart. The uppermost normal is correlated to the Brunhes chron (<780 ka), the immediately underlying normal-reverse represents Jaramillo and latest Matuyama time (0.99–0.78 Ma), while the lowest reversely magnetized sediment is certainly pre-Jaramillo (>1.07 Ma) in age. On the basis of sedimentation rates, the base of the Midnight Dome loess may be 1.4–1.5 Ma in age. This suggests an age range broadly similar to the intermediate terrace age at Archibald's Terrace from the Mosquito Gulch tephra at 1.22 ± 0.49 Ma (±2σ; Naeser et al. 1982).

Origin and extent of the paleomagnetic record

Since the stratigraphic units occur in terraces, their topographic position allows them to be related across valleys in the local area, their stratigraphic position can be related with confidence, and a composite stratigraphy developed (Fig. 11). At least 8 changes in polarity are noted from the earliest preglacial Lower White Channel to Brunhes-age normal sediment in the Midnight Dome loess. Two possible interpretations are considered in the correlation with the geomagnetic polarity timescale, based on whether the Klondike gravel – Upper White Channel normal polarity zone corresponds with the Olduvai normal subchron or, al-

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Fig. 11. Composite stratigraphy of the two main Klondike valley terraces: Australia and Jackson hills (uppermost terrace) and Midnight Dome Terrace (intermediate). Bold N, R represent normal and reverse magnetization, respectively.



ternatively, the Gauss Normal chron, termed hypotheses A and B, respectively, (Fig. 12). Both scenarios assume that the Cobb Mountain (1.201–1.211 Ma) and Réunion (2.14–2.15 Ma) subchrons are too short to have been recorded by these two units.

Hypothesis A: Olduvai first glaciation

Hypothesis A proposes that the normal polarity represented by the Klondike gravel and Upper White Channel gravel corresponds with the Olduvai normal subchron (Fig. 12). The implications of this correlation are (1) White Channel sedimentation must at least date to the mid-Gauss chron (Kaena 3.04–3.11 Ma); (2) the early Matuyama (pre-Olduvai) is recorded by the upper 4 m of reversely magnetized sediment at Jackson Hill; and (3) the earliest evidence of periglacial conditions in the Upper White Channel gravel likely occurs in reversely magnetized sediment, which is ascribed to the early Matuyama (1.95–2.58 Ma).

Hypothesis B: pre-Olduvai first glaciation

Hypothesis B proposes that the normal polarity of the Klondike gravel and Upper White Channel represents deposition during the Gauss chron (>2.58 Ma). This correlation assumes that the Cobb Mountain, Olduvai, and Réunion subchrons are not preserved (Fig. 12). The corollaries of this proposal are (1) the first Cordilleran Ice Sheet advance into western Yukon occurred by 2.58 Ma; (2) deposition of Lower White Channel gravel (R-N composite between North Australia and Jackson hills), was initiated prior to the Mammoth subchron (>3.22 Ma), and probably earlier (Gilbert chron); and, (3) the earliest evidence of periglacial conditions occurs most likely by the Kaena subchron (3.04–3.11 Ma).

Preferred interpretation of the paleomagnetic record

Both hypotheses A and B indicate that Lower White Channel gravel was deposited by at least the Gauss chron (>2.58 Ma). However, the assignment of a Gauss versus Olduvai age for the Upper White Channel and Klondike gravel units is difficult on the basis of paleomagnetic data alone. A Gauss age for the Upper White Channel – Klondike gravel magnetopolarity zone is preferred for the following reasons.

(1) Evidence from elsewhere in the northern Cordillera (Westgate et al. 1990; Duk-Rodkin et al. 1996; Barendregt et al. 1996), indicates that full-glacial conditions likely existed in the northern Cordillera prior to the Olduvai (>1.77 Ma). Near Fairbanks, Alaska, Gold Hill Loess deposition, resulting from advancing glaciers in the Tanana River valley began by the Mammoth subchron (ca. 3 Ma; Westgate et al. 1990).

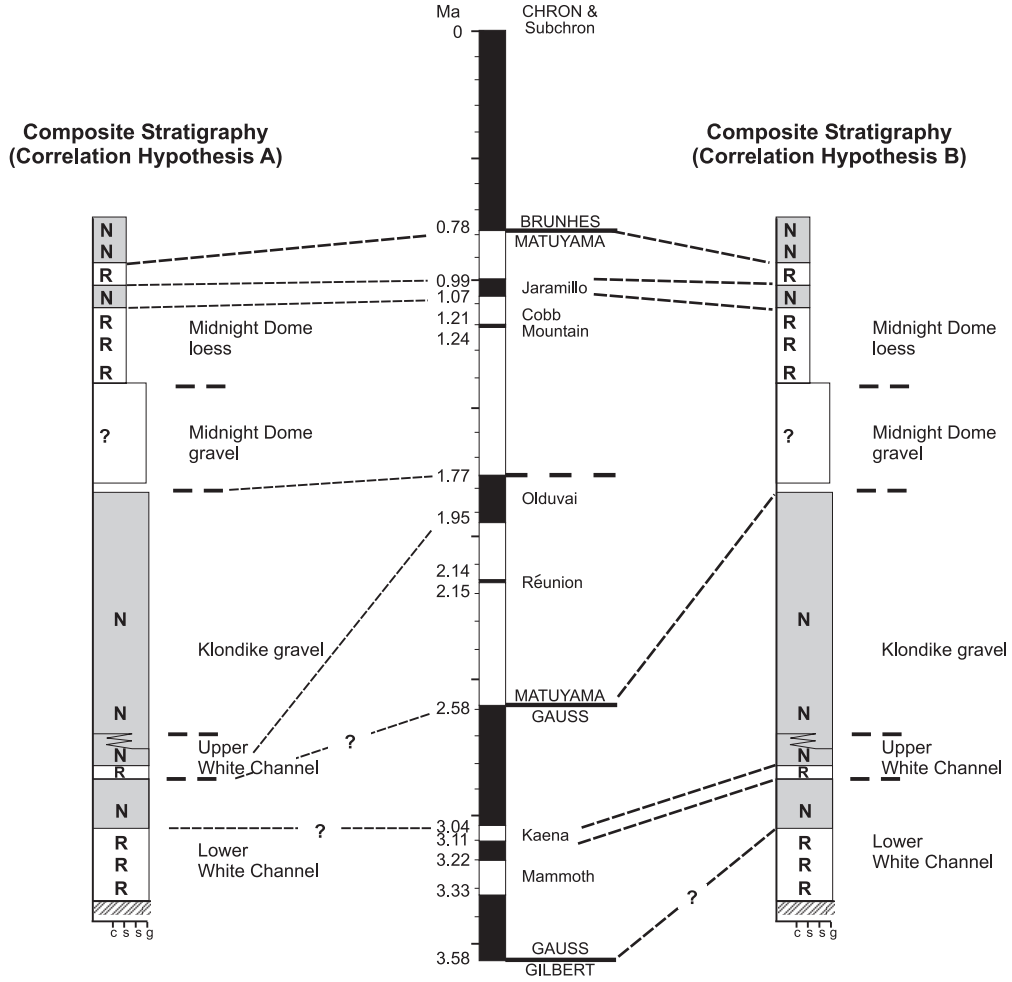
(2) Cores from the North Pacific Ocean indicate large-scale glaciation began abruptly in the Northern Hemisphere during the Gauss chron (before 2.58 Ma; Krissek 1995; Rea et al. 1995).

(3) Regional evidence from the Tintina Trench north of the Dawson area (Duk-Rodkin and Barendregt 1997) also reveals a normal-reverse-normal-reverse polarity sequence of early glacial sediments, arguably representing late Gauss to early Matuyama deposition (suggested as Gauss through late Matuyama glacial events).

(4) The onset of permafrost in the Klondike record occurs in close association with the first direct evidence of glaciation (Upper White Channel gravel which interfingers with the Klondike gravel). In other areas of the northern Cordillera, pre-Olduvai permafrost has been documented in the Gauss chron (Péwé 1989; Westgate et al. 1990; Repenning and Brouwers 1992). Therefore, it seems reasonable that the

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Fig. 12. Correlation of Klondike valley stratigraphy to the geomagnetic polarity timescale based on hypothesis A: Olduvai age for Upper White Channel gravel – Klondike Wash; and hypothesis B: Gauss age (preferred interpretation) for Upper White Channel gravel – Klondike gravel.



first ice-wedge casts in the Upper White Channel are also of Gauss age.

Evidence of Late Cenozoic environmental change

Pollen in the Lower White Channel is generally poorly preserved and no macrofossils have been confirmed from the gravel deposits. Eight samples were processed for pollen analyses from overbank deposits, but only one yielded a countable sum. White (1996) identified a nonglacial assemblage from a sample midway through the Jackson Hill exposure. The assemblage was dominated by coniferous trees, *Picea, Pinus,* with rare *Abies* grains, suggesting dense boreal forest conditions with little understory vegetation in the area (White 1996). This nonglacial assemblage is similar to preglacial deposits (mid-Pliocene) from the Circle area and the Nenana gravel in adjacent Alaska (Ager et al. 1994). In particular, *Pinus* is found in preglacial material in Alaska and Yukon, but not in subsequent interglacial deposits so far characterized in Yukon–Alaska (Schweger et al. 1999). This pollen data is consistent with previous observations of the Lower White Channel being of preglacial age (McConnell 1905, 1907; Hughes et al. 1972; Morison 1985).

Paleomagnetic and stratigraphic data, based on the preferred correlation to the geomagnetic polarity time scale, suggest that a major climatic change from preglacial conditions to alternating glacial–interglacial conditions occurred in the Late Pliocene, during the Gauss chron, recorded by the Klondike gravel (>2.58 Ma). Evidence of a slightly earlier Late Pliocene periglacial climate is found in the Upper White Channel gravel (ice-wedge casts), suggesting that permafrost was likely present in the area by about 3 Ma (Kaena or Mammoth subchrons).

The Klondike gravel marks the first direct evidence of glaciation in the area with the deposition by the Cordilleran Ice Sheet of extra-basinal clasts from north of Tintina Trench. In contrast to other Late Pliocene – Early Pleistocene records in the northern Cordillera which show a reversely magnetized (Matuyama) earliest glacial record (Duk-Rodkin et al. 1996; Barendregt et al. 1996; Jackson et al. 1996), the preglacial to glacial transition of the Klondike valley occurred during a normal interval (assigned to the Gauss chron >2.58 Ma), suggesting a pre-Matuyama age for the earliest Cordilleran Ice Sheet. If a southerly flowing preglacial Yu-

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kon River was diverted to its present location by Late Cenozoic ice as suggested (Tempelman-Kluit 1980; Duk-Rodkin 1997), it must have occurred by this time. Subsequent to deposition of the Klondike gravel, the long hiatus in the record may be a result of an extended period of erosion, perhaps as much as 1 Ma, during which time Klondike River valley was incised by about 100 m.

The Midnight Dome loess sequence records at least four cycles of sedimentation under glacial and interglacial conditions based on pollen assemblages and sedimentary facies (Froese and Schweger 1999; Schweger et al. 1999). In general, this stratigraphic record of loess and reworked loess suggests deposition of loess and hillslope facies in response to cold (periglacial or glacial) conditions with the transition of upland boreal forest vegetation to steppe tundra. During interglacials, there appears to be negligible deposition when the boreal forest was reestablished in the area. Major colluvial diamicts (>0.2 m) can be traced laterally tens to hundreds of metres across exposures and tend to underlie massive silt units (primary loess). It is assumed that the major massive silt units are primary loess and their deposition was conditioned by ice in the headwaters of Yukon and Klondike rivers as with Late Wisconsinan loess deposition in the area (Fraser and Burn 1997).

Conclusions

Magnetostratigraphic investigation of alluvial, glaciofluvial, and eolian deposits in the lower Klondike River valley has allowed correlation of preglacial, glacial, and interglacial sediments to the geomagnetic polarity time scale over the last 3 Ma.

During the Early to mid-Pliocene, nonglacial conditions prevailed, as recorded by the Lower White Channel gravel. Deterioration of climate in the Late Pliocene resulted in a drop in average annual temperature such that permafrost was probably present in the region by about 3 Ma. Increased sediment contribution to the nonglaciated valleys resulted in aggradation of the valleys and local preservation of ice-wedge casts in the alluvial deposits of the Upper White Channel gravel. Outwash gravel deposits of a Late Pliocene Cordilleran Ice Sheet, the Klondike gravel event before 2.58 Ma (Gauss normal chron), interfinger with normally magnetized Upper White Channel deposits indicating a close association with the onset of periglacial conditions in the northern Cordillera and the growth of the Cordilleran Ice Sheet.

Incision of the Klondike River valley by 100 m following deposition of the Klondike gravel caused an hiatus in the preserved record of about 1 Ma. A depositional history of a pre-Reid interglacial paleo-Klondike River, and the onset of a pre-Reid glaciation in the Ogilvie Mountains is recorded by the Midnight Dome gravel. This gravel unit is overlain by up to 15 m of loess and hillslope deposits, the Midnight Dome loess, which record four magnetic polarity zones (R-N-R-N) and at least three interglacials from approximately 1.5 Ma to <780 ka, including the Jaramillo subchron (1.1 Ma).

Sediments in Klondike River valley provide a record of multiple glaciation and interglaciation prior to Reid glaciation (>200 ka), and represent one of the oldest records of Plio–Pleistocene climate change in North America.

Acknowledgments

The cooperation and assistance of Klondike area placer miners in completing this study is gratefully acknowledged. In particular, Torfinn Djukastein, Walter Hinnek, Brendan White, Frank Short, Tony Beets, and Bob Cattermole went out of their way to help. Many other miners and their families made this study feasible and a pleasure to carry out. Aaron Clack, Anne Williams, and Scott Barnes are thanked for their enthusiastic support in fieldwork. Jeff Bond, Chris Burn, Alejandra Duk-Rodkin, Fran Hein, Bill Lebarge, and Scott Smith provided useful discussions in the field. Charles Schweger and James White expeditiously completed the pollen analyses for this study. Alejandra Duk-Rodkin, Colm O'Cofaigh, Fran Hein, and Derald Smith provided valuable discussion of an earlier version of this manuscript. The comments and suggestions by reviewers John Clague and Tom Ager substantially improved the manuscript. This work was supported by Natural Sciences and Engineering Research Council of Canada grants to R.W. Barendregt, F.J. Hein, and D.G. Smith; Northern Science Training Program grants from the Department of Indian Affairs and Northern Development, the Terrain Sciences Division, Geological Survey of Canada, and the Yukon Geology Program, Whitehorse.

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